

EE 446: TinyML

Lab 2: TinyML Hardware and Sensor Setup

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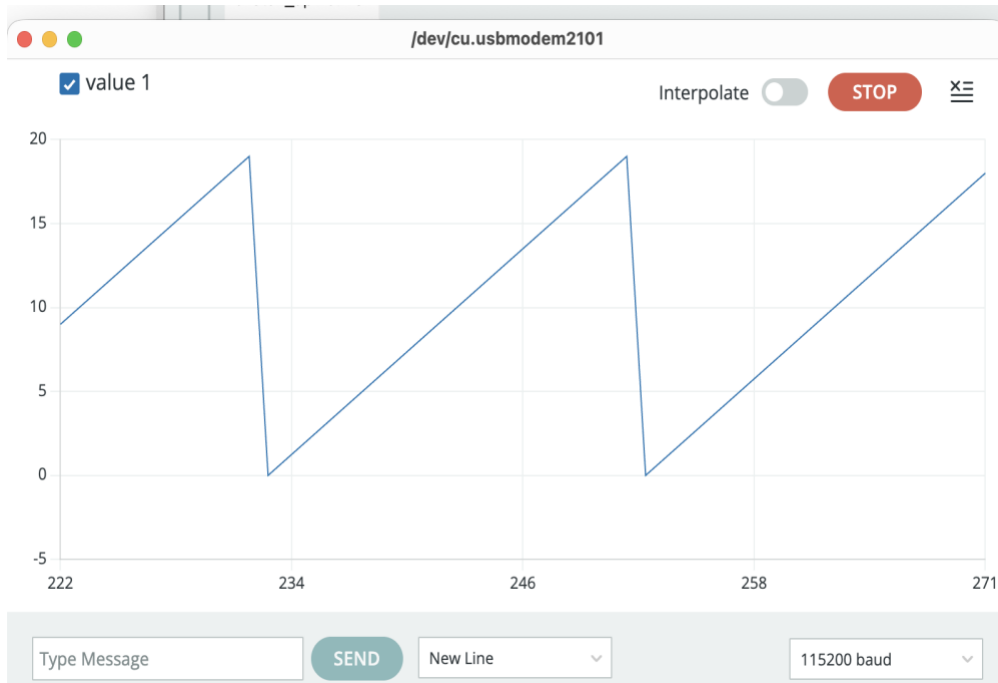
Student ID: 2262052

GitHub Link: <https://github.com/asafij1111/TinyML/tree/main/Lab2>

Disclaimer: I used Claude co-work to organize my screenshots in this pdf but all answers are my own. I used Claude to comment and format my code but all logic and functional code was written by me.

Task 4 — Serial Monitor / Serial Plotter

Screenshot of Serial Monitor showing the counter output:



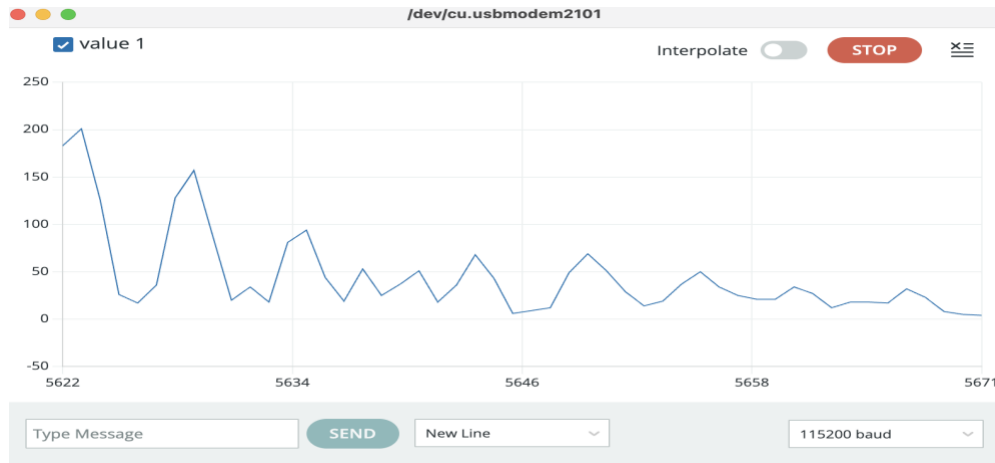
Screenshot of Serial Plotter showing the live trace:

The screenshot shows the Serial Monitor window with the title bar indicating the device path `/dev/cu.usbmodem2101`. The output area displays a live trace of the counter output, showing a series of timestamps and counts. The trace is as follows:

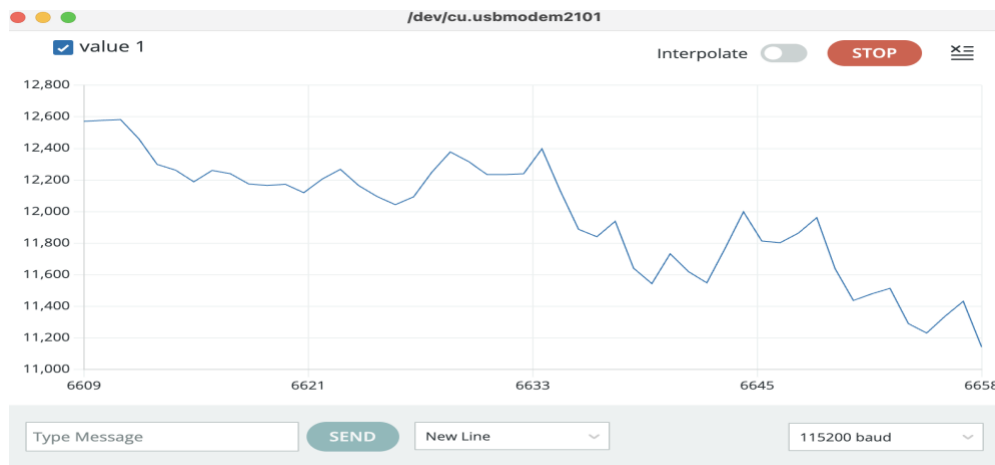
Timestamp	Count
17:41:34.599	26
17:41:35.600	27
17:41:36.597	28
17:41:37.598	29
17:41:38.600	30
17:41:39.598	31
17:41:40.594	32
17:41:41.625	33
17:41:42.623	34
17:41:43.619	35
17:41:44.595	36
17:41:45.592	37
17:41:46.594	38

Task 5 — Microphone Verification

Screenshot of Serial Plotter while the environment is relatively quiet:



Screenshot of Serial Plotter while speaking, clapping, or tapping near the board:

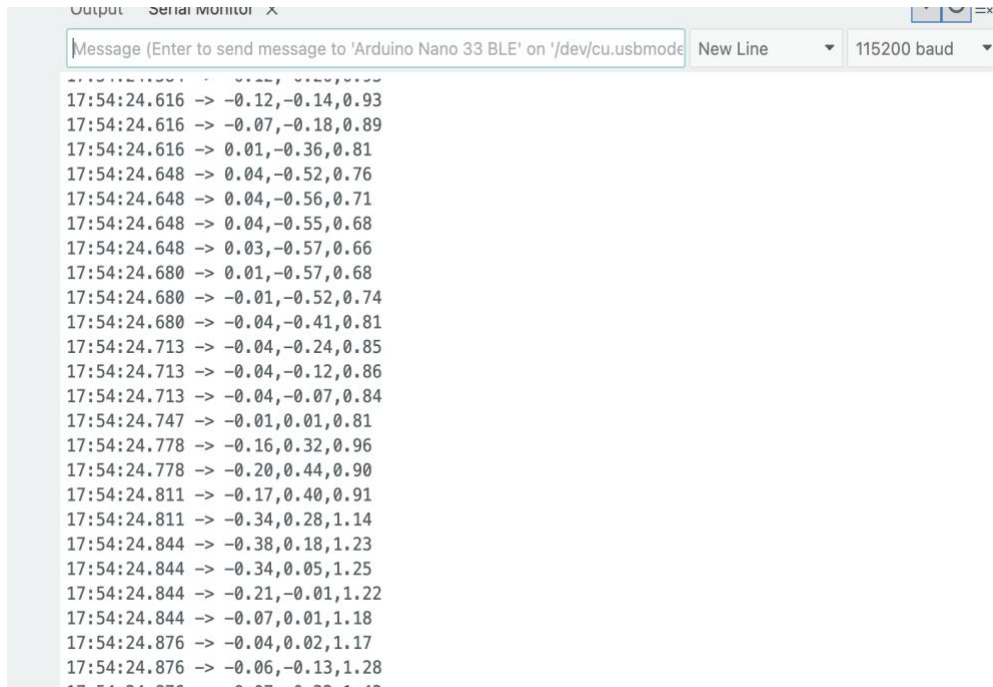


One short sentence describing what action caused the clearest change in the plotted signal:

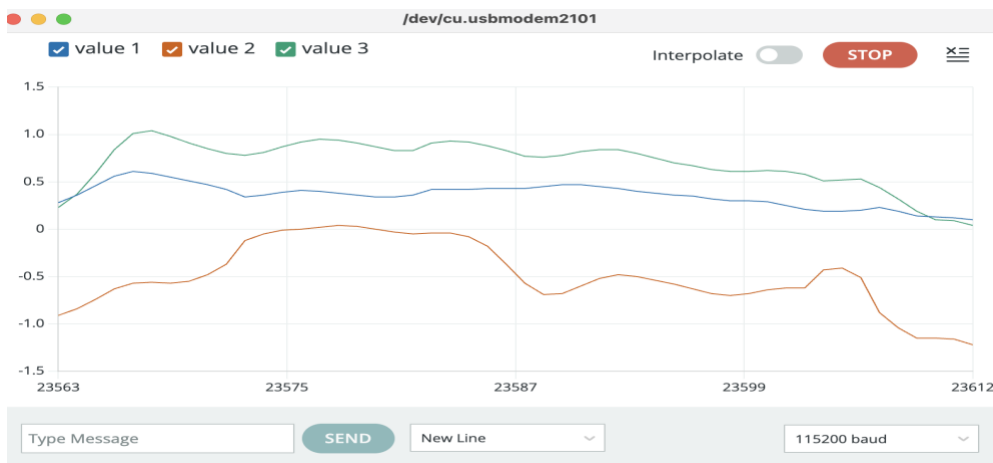
Loudly humming or singing with the nano close to my mouth was the loudest and resulted in the high value of above 10,000.

Task 6A — IMU Verification (Accelerometer)

Screenshot of Serial Monitor showing changing accelerometer values:



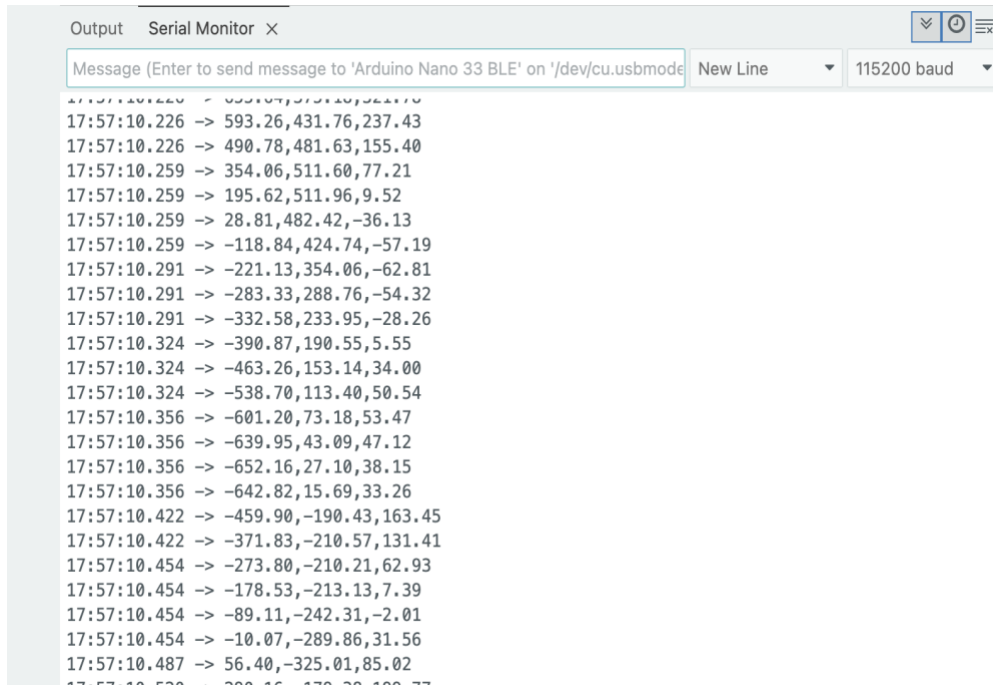
Screenshot of Serial Plotter showing the three changing channels:



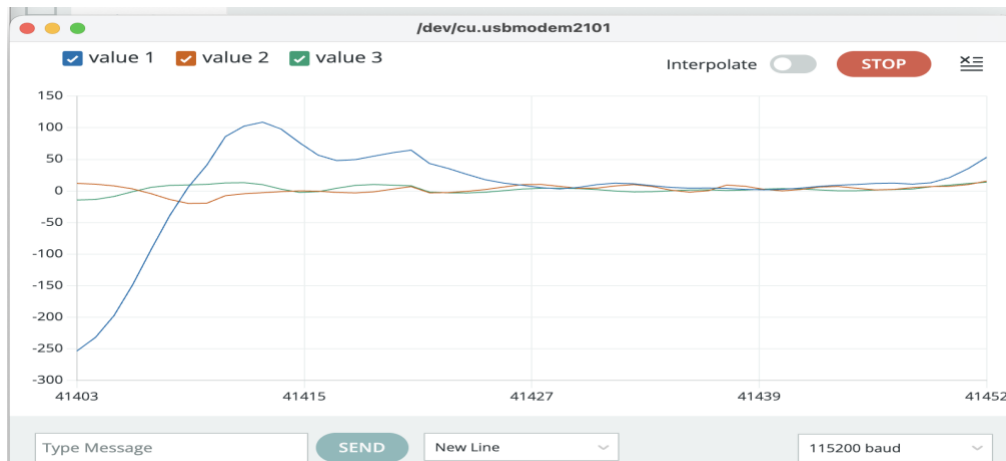
One short sentence explaining which movement produced the clearest change:
Shaking the Nano rapidly created the most substantial change.

Task 6B — Gyroscope Verification

Screenshot of Serial Monitor showing changing gyroscope values:



Screenshot of Serial Plotter showing the three changing channels:



One short sentence describing which rotation produced the clearest change:

The most change came from rotating quickly along the axis of the port.

Task 6C — Magnetometer Verification

Screenshot of Serial Monitor showing changing magnetometer values:

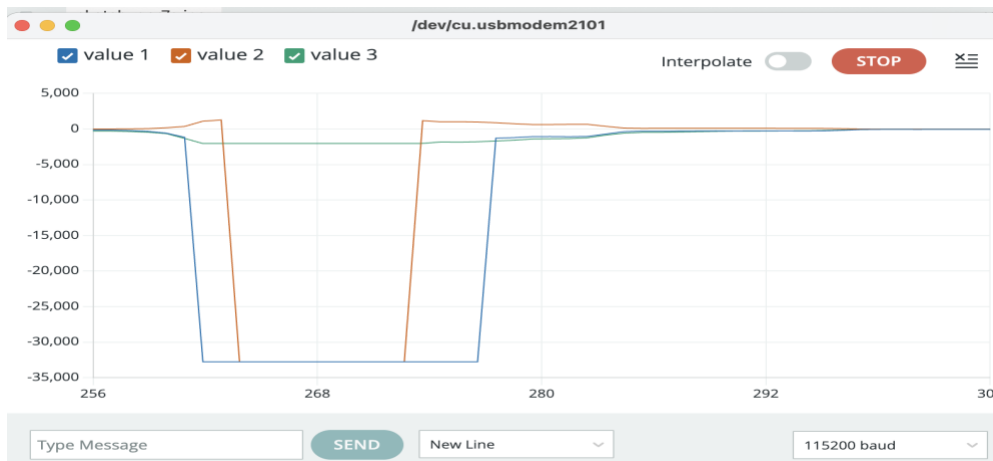
```
20
21 // Print comma-separated values
22 Serial.print(x);
23 Serial.print(",");
24 Serial.print(y);
25 Serial.print(",");
26 Serial.println(z);
27 }
28 }
```

Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmode New Line 115200 baud

-41.00,-12.00,-49.00
-32.00,-17.00,-2.00
-32.00,-24.00,13.00
-31.00,-19.00,7.00
-34.00,-17.00,-11.00
-33.00,-19.00,-4.00
-30.00,-20.00,12.00
-30.00,-23.00,12.00
-31.00,-23.00,10.00
-32.00,-16.00,0.00
-31.00,-20.00,8.00
-34.00,-24.00,17.00
-31.00,-22.00,10.00
-34.00,-18.00,-

Screenshot of Serial Plotter showing the three changing channels:

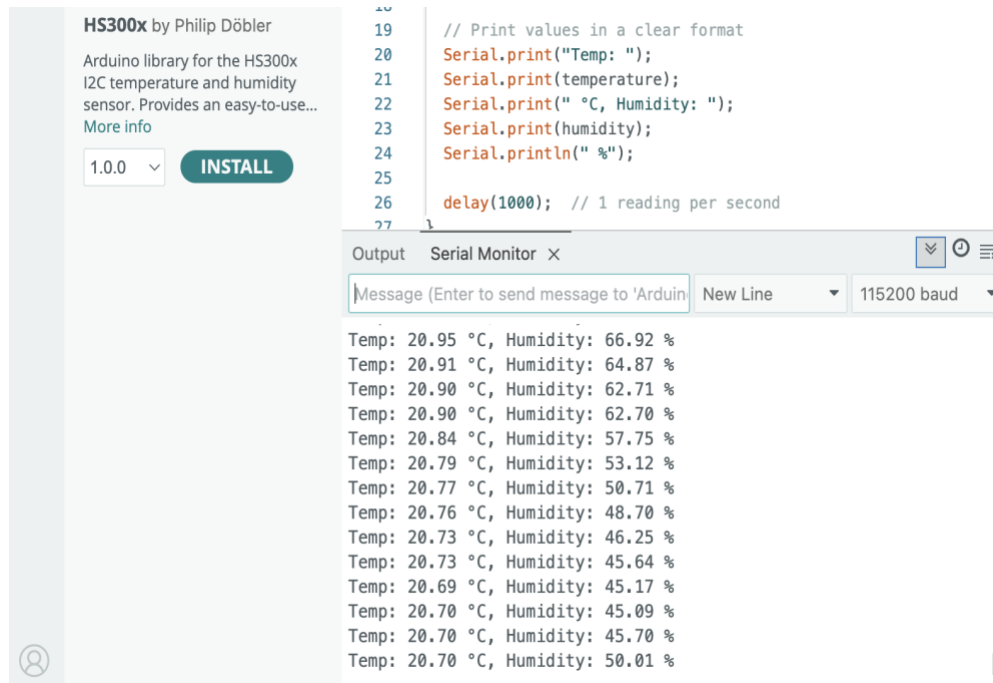


One short sentence describing what action produced the clearest change:

Putting the nano close to my computer battery made the largest different and resulted in the massive jump in the mag value.

Task 7 — Humidity and Temperature Verification

Screenshot of Serial Monitor showing stable humidity and temperature values:



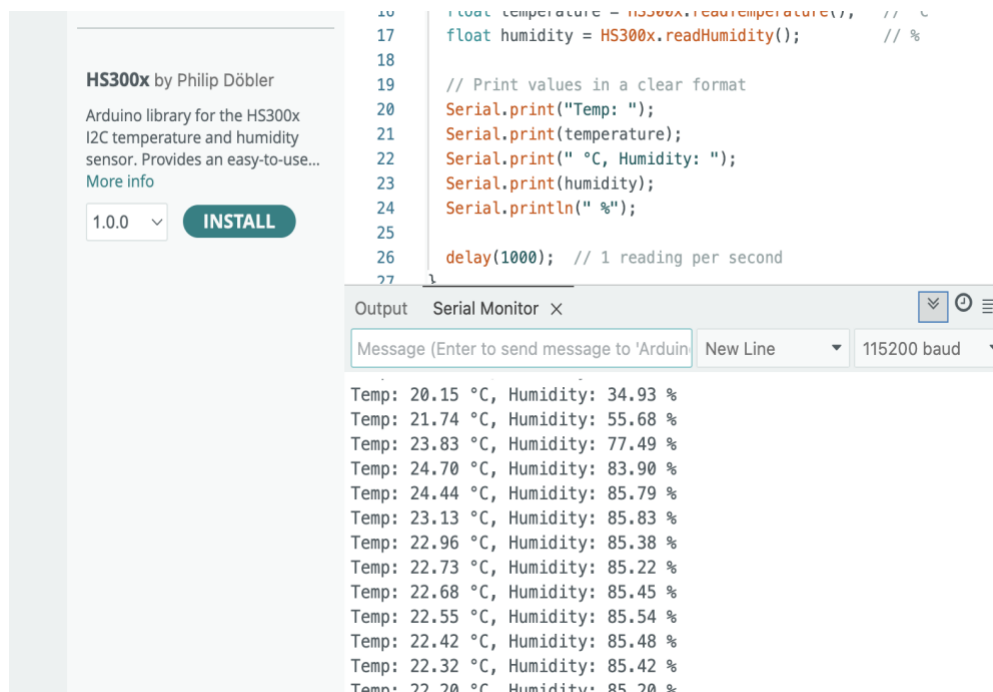
The screenshot shows the Arduino IDE interface. On the left, the 'HS300x' library by Philip Döbler is listed with version 1.0.0 and an 'INSTALL' button. The main editor displays the following code:

```
19 // Print values in a clear format
20 Serial.print("Temp: ");
21 Serial.print(temperature);
22 Serial.print(" °C, Humidity: ");
23 Serial.print(humidity);
24 Serial.println(" %");
25
26 delay(1000); // 1 reading per second
27 }
```

The Serial Monitor window at the bottom shows the following output:

```
Temp: 20.95 °C, Humidity: 66.92 %
Temp: 20.91 °C, Humidity: 64.87 %
Temp: 20.90 °C, Humidity: 62.71 %
Temp: 20.90 °C, Humidity: 62.70 %
Temp: 20.84 °C, Humidity: 57.75 %
Temp: 20.79 °C, Humidity: 53.12 %
Temp: 20.77 °C, Humidity: 50.71 %
Temp: 20.76 °C, Humidity: 48.70 %
Temp: 20.73 °C, Humidity: 46.25 %
Temp: 20.73 °C, Humidity: 45.64 %
Temp: 20.69 °C, Humidity: 45.17 %
Temp: 20.70 °C, Humidity: 45.09 %
Temp: 20.70 °C, Humidity: 45.70 %
Temp: 20.70 °C, Humidity: 50.01 %
```

Screenshot showing changed values after introducing a nearby hand or gentle breath:



The screenshot shows the same Arduino IDE interface as the previous one. The code is identical. The Serial Monitor window shows the following output:

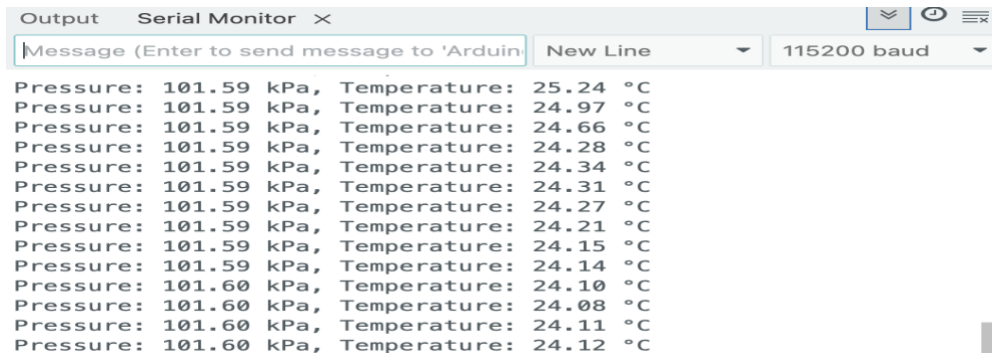
```
Temp: 20.15 °C, Humidity: 34.93 %
Temp: 21.74 °C, Humidity: 55.68 %
Temp: 23.83 °C, Humidity: 77.49 %
Temp: 24.70 °C, Humidity: 83.90 %
Temp: 24.44 °C, Humidity: 85.79 %
Temp: 23.13 °C, Humidity: 85.83 %
Temp: 22.96 °C, Humidity: 85.38 %
Temp: 22.73 °C, Humidity: 85.22 %
Temp: 22.68 °C, Humidity: 85.45 %
Temp: 22.55 °C, Humidity: 85.54 %
Temp: 22.42 °C, Humidity: 85.48 %
Temp: 22.32 °C, Humidity: 85.42 %
Temp: 22.20 °C, Humidity: 85.20 %
```

One short sentence describing what action produced the clearest change:

Humidity changed a lot when I put my hand on the nano completely covering it. It went up a lot because my hands were sweaty. Temp went up the most from breathing hot air on the sensors.

Task 8 — Barometric Pressure Verification

Screenshot of Serial Monitor showing pressure and temperature values:

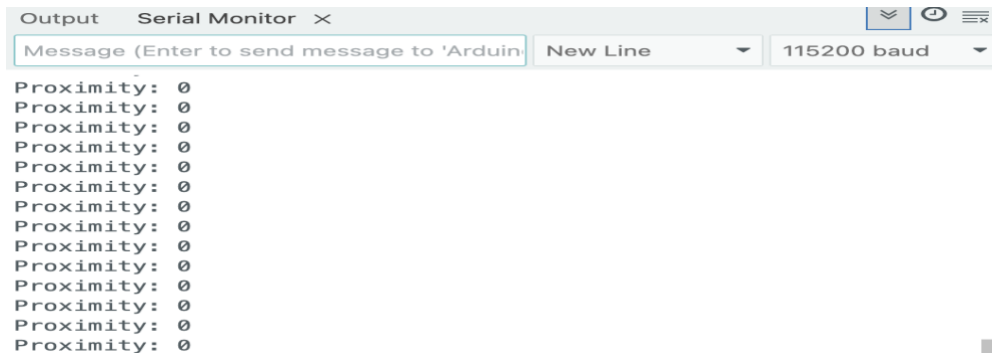


One short sentence stating whether the pressure reading was mostly stable or changed slightly during testing:

The pressure was extremely stable when still but could be changed by swinging the Nano or breathing on it.

Task 9A — Proximity Verification

Screenshot of Serial Monitor with the hand far from the board:



Screenshot of Serial Monitor with the hand close to the board:

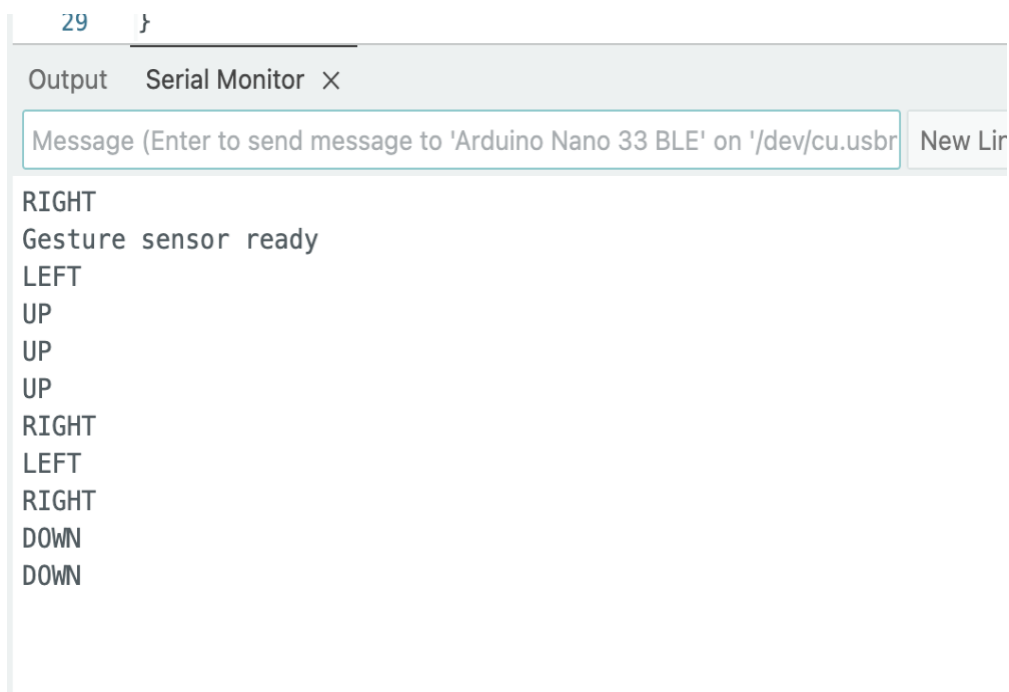


One short sentence describing how the value changed:

The value would be high from about a foot away but then rapidly go towards zero as my hand got within around a foot. From within 1-2 inches the value was 0 or less than 10.

Task 9B — Gesture Verification

Screenshot showing a recognized gesture in Serial Monitor:

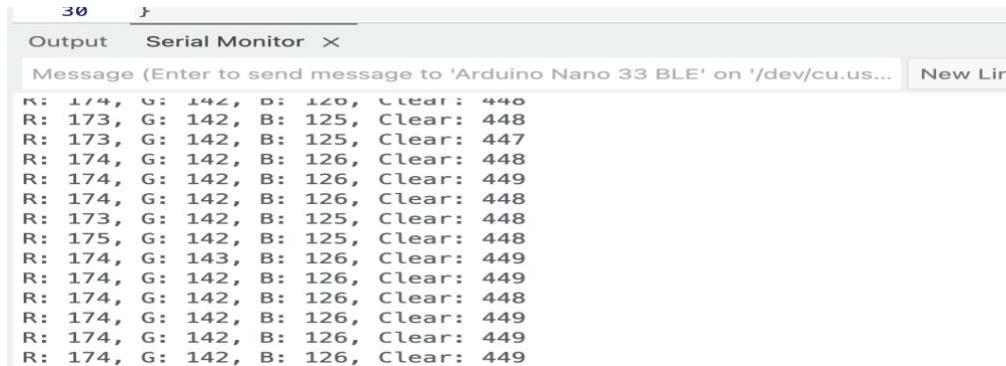


One short sentence stating which gesture was recognized most reliably:

All the gestures were relatively similar reliability. The most important factor was moving slowly enough and at the correct distance of about a foot away.

Task 9C — Ambient Light and RGB Color Verification

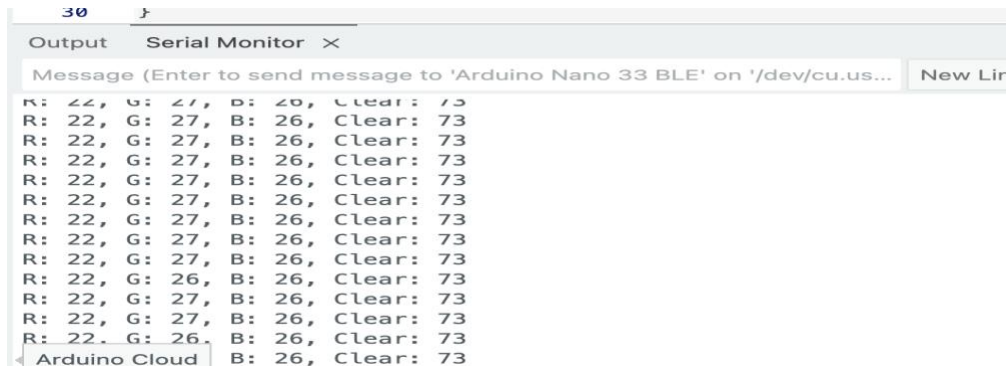
Screenshot of Serial Monitor under baseline room lighting:



A screenshot of the Arduino IDE Serial Monitor window. The window title is "30" and it has a close button. The "Output" tab is selected. The text area shows a series of 15 lines of data, each representing an RGB color reading and a "Clear" value. The data is as follows:

R	G	B	Clear
174	142	126	448
173	142	125	448
173	142	125	447
174	142	126	448
174	142	126	449
174	142	126	448
173	142	125	448
175	142	125	448
174	143	126	449
174	142	126	449
174	142	126	448
174	142	126	449
174	142	126	449
174	142	126	449
174	142	126	449

Screenshot of Serial Monitor after changing lighting conditions or presenting a colored object:



A screenshot of the Arduino IDE Serial Monitor window, similar to the first one. The "Output" tab is selected. The text area shows a series of 15 lines of data, each representing an RGB color reading and a "Clear" value. The data is as follows:

R	G	B	Clear
22	27	26	73
22	27	26	73
22	27	26	73
22	27	26	73
22	27	26	73
22	27	26	73
22	27	26	73
22	27	26	73
22	27	26	73
22	26	26	73
22	27	26	73
22	27	26	73
22	27	26	73
22	26	26	73
22	26	26	73

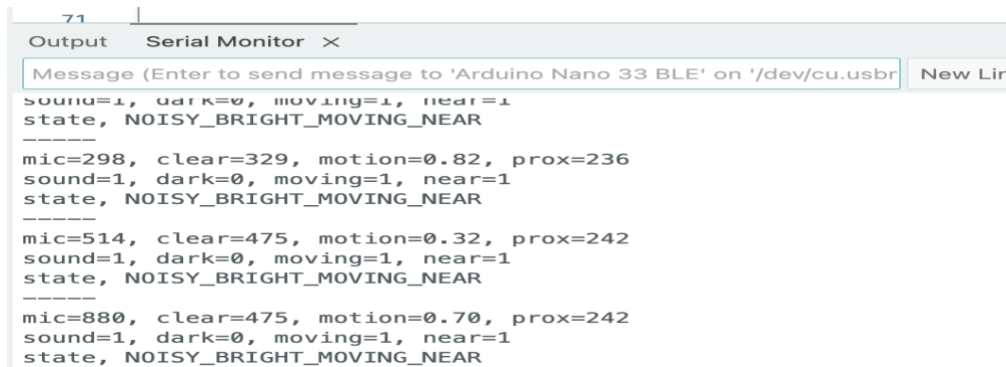
One short sentence describing which change produced the clearest difference:

Completely covering the Nano with my hand made a big difference but also putting my purple iPhone in front of it made a noticeable difference as well which is what is shown in the second image.

Task 10 — Smart Workspace Situation Classifier

GitHub link to Arduino code:

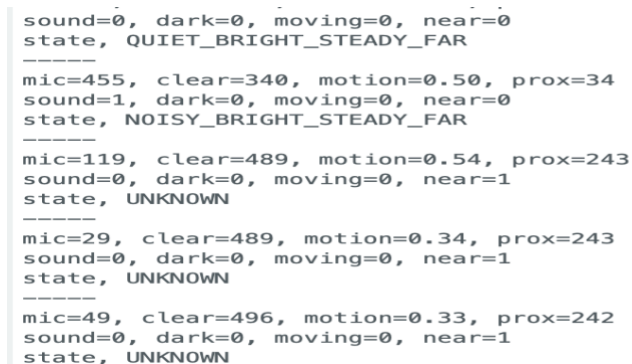
Screenshot for *NOISY_BRIGHT_MOVING_NEAR*:



The screenshot shows the Arduino Serial Monitor interface. The title bar indicates the connection is to 'Arduino Nano 33 BLE' on '/dev/cu.usbr'. The output window displays the following data:

```
71 |  
Output Serial Monitor X  
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbr') New Lir  
sound=1, dark=0, moving=1, near=1  
state, NOISY_BRIGHT_MOVING_NEAR  
-----  
mic=298, clear=329, motion=0.82, prox=236  
sound=1, dark=0, moving=1, near=1  
state, NOISY_BRIGHT_MOVING_NEAR  
-----  
mic=514, clear=475, motion=0.32, prox=242  
sound=1, dark=0, moving=1, near=1  
state, NOISY_BRIGHT_MOVING_NEAR  
-----  
mic=880, clear=475, motion=0.70, prox=242  
sound=1, dark=0, moving=1, near=1  
state, NOISY_BRIGHT_MOVING_NEAR
```

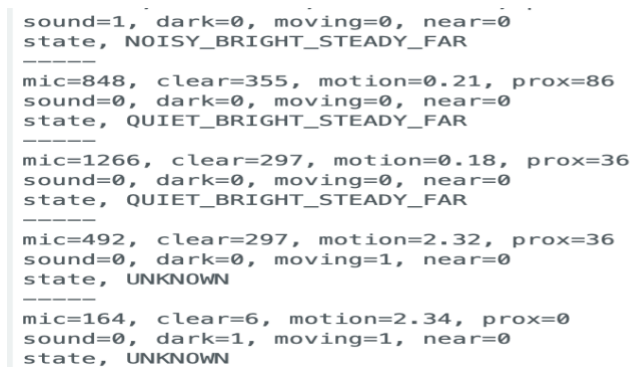
Screenshot for *QUIET_BRIGHT_STEADY_FAR*:



The screenshot shows the Arduino Serial Monitor interface displaying the following data:

```
sound=0, dark=0, moving=0, near=0  
state, QUIET_BRIGHT_STEADY_FAR  
-----  
mic=455, clear=340, motion=0.50, prox=34  
sound=1, dark=0, moving=0, near=0  
state, NOISY_BRIGHT_STEADY_FAR  
-----  
mic=119, clear=489, motion=0.54, prox=243  
sound=0, dark=0, moving=0, near=1  
state, UNKNOWN  
-----  
mic=29, clear=489, motion=0.34, prox=243  
sound=0, dark=0, moving=0, near=1  
state, UNKNOWN  
-----  
mic=49, clear=496, motion=0.33, prox=242  
sound=0, dark=0, moving=0, near=1  
state, UNKNOWN
```

Screenshot for *NOISY_BRIGHT_STEADY_FAR*:



The screenshot shows the Arduino Serial Monitor interface displaying the following data:

```
sound=1, dark=0, moving=0, near=0  
state, NOISY_BRIGHT_STEADY_FAR  
-----  
mic=848, clear=355, motion=0.21, prox=86  
sound=0, dark=0, moving=0, near=0  
state, QUIET_BRIGHT_STEADY_FAR  
-----  
mic=1266, clear=297, motion=0.18, prox=36  
sound=0, dark=0, moving=0, near=0  
state, QUIET_BRIGHT_STEADY_FAR  
-----  
mic=492, clear=297, motion=2.32, prox=36  
sound=0, dark=0, moving=1, near=0  
state, UNKNOWN  
-----  
mic=164, clear=6, motion=2.34, prox=0  
sound=0, dark=1, moving=1, near=0  
state, UNKNOWN
```

Screenshot for *QUIET_DARK_STEADY_NEAR*:

```
sound=0, dark=1, moving=0, near=1
state, QUIET_DARK_STEADY_NEAR
-----
mic=342, clear=6, motion=0.17, prox=0
sound=0, dark=1, moving=0, near=1
state, QUIET_DARK_STEADY_NEAR
-----
mic=543, clear=8, motion=0.16, prox=0
sound=0, dark=1, moving=0, near=1
state, QUIET_DARK_STEADY_NEAR
-----
mic=307, clear=8, motion=0.18, prox=0
sound=0, dark=1, moving=0, near=1
state, QUIET_DARK_STEADY_NEAR
-----
mic=49, clear=18, motion=0.20, prox=0
sound=0, dark=1, moving=0, near=1
state, QUIET_DARK_STEADY_NEAR
```

Short explanation of how you selected your thresholds:

I basically looked at each of the 4 categories outputs and put them at extremes such as up to the window vs covered from the light. I chose the threshold value as somewhere in the middle of the two. This way it was easy to clearly show one or the other. The values are all very different because of the way the different sensor outputs are formatted.

Task 11 — Environment Event Detector

GitHub link to Arduino code:

Screenshot for BASELINE_NORMAL:

```
rh=66.68 temp=31.72 mag=36.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
-----
rh=54.26 temp=31.31 mag=74.00 clear=3029
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
-----
rh=53.46 temp=31.46 mag=72.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
-----
rh=73.78 temp=31.86 mag=67.00 clear=23
humid_jump=1 temp_rise=0 mag_shift=0 light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
-----
rh=75.36 temp=31.12 mag=59.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
```

Screenshot for BREATH_OR_WARM_AIR_EVENT:

```
rh=22.00 temp=31.32 mag=59.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
-----
rh=38.89 temp=31.36 mag=33.00 clear=0
humid_jump=1 temp_rise=0 mag_shift=0 light_or_color_change=0
-----
rh=61.91 temp=31.58 mag=32.00 clear=0
humid_jump=1 temp_rise=0 mag_shift=0 light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
-----
rh=69.57 temp=31.68 mag=33.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
-----
rh=72.82 temp=31.73 mag=27.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
```

Screenshot for LIGHT_OR_COLOR_CHANGE_EVENT:

```
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
-----
rh=21.87 temp=31.39 mag=39.00 clear=4097
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
-----
rh=21.64 temp=31.37 mag=38.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
-----
rh=21.39 temp=31.36 mag=42.00 clear=4097
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
-----
rh=20.99 temp=31.36 mag=36.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
-----
rh=20.64 temp=31.33 mag=44.00 clear=4097
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
```

Screenshot for MAGNETIC_DISTURBANCE_EVENT:

```
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
-----
rh=23.93 temp=31.60 mag=75.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
-----
rh=23.07 temp=31.51 mag=215.00 clear=4097
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
-----
rh=22.95 temp=31.47 mag=1106.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
-----
rh=23.50 temp=31.46 mag=33455.00 clear=4097
humid_jump=0 temp_rise=0 mag_shift=1 light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
-----
rh=22.52 temp=31.43 mag=33072.00 clear=0
humid_jump=0 temp_rise=0 mag_shift=0 light_or_color_change=1
```

Short explanation of how you selected your thresholds and cooldown logic:

Similar to task 10 I chose my thresholds so that when I didn't intend a change it would stay stable. I first looked at the outputs of each of the values and did an action to try to move it a lot. I then made the threshold a difference value that is less than that extreme change but more than the stable state jitter. I used a global cooldown of 1 second delay so that the output is very readable and there is enough time for the values to change meaningfully while I manipulate the sensor inputs.
